

Morse Lemma. (Normal forms) differential topology

Diffeomorphisms 微分同胚

光滑: 无限可微

enables one to analyze the topology of a manifold by studying differentiate functions on that manifold.

定义: bijective mapping: $F: U \rightarrow V$, F^{-1} denoting the inverse mapping.
 F is C^k -diffeomorphism ($k \geq 1$) if $F \in C^k(U, V)$ and $F^{-1} \in C^k(V, U)$
 $\downarrow$
 Continuous differentiable. C^1 : 一阶连续可微. C^0 : 连续.

Optimality Conditions 最优性条件

优化问题中: 如果 x^* 是局部最优解
 它必须满足什么条件?

本质: "一个解是最优的, 需要满足某些方程/条件"

极值问题的必要条件
 拉格朗日乘子法

- morphism: 映射式结构

isomorphism 同构
 homeomorphism 同胚
 diffeomorphism 微分同胚

思想: calculus of variations 变分法

the method of Lagrange multipliers: 寻找 local maxima and minima.
 (Multivariable Calculus)

例子: maximize $f(x, y) = xy$
 Subject to $3x + y = 720$

$$g(x, y) = 3x + y - 720 = 0$$

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y) = xy + \lambda(3x + y - 720)$$

$$F_x(x, y, \lambda) = 0 \Rightarrow y + 3\lambda = 0$$

$$F_y(x, y, \lambda) = 0 \Rightarrow x + \lambda = 0$$

$$F_\lambda(x, y, \lambda) = 0 \Rightarrow 3x + y - 720 = 0$$

$$\Rightarrow \lambda = -120 \quad \begin{matrix} x = 120 \\ y = 360 \end{matrix}$$

The solutions are critical points of F .

PCA 优化最大方差: $\max u^T S u$

Subject to $\|u\| = 1$

$$F(u, \lambda) = u^T S u + \lambda(u^T u - 1)$$

$$F_u = u^T S + \lambda u^T = 0$$

$$F_\lambda = u^T u - 1 = 0$$

$$u^T S = -\lambda u^T \Rightarrow S^T u = -\lambda u$$

$\rightarrow$ eigenvalue.

$$\text{对称} \Rightarrow S u = -\lambda u$$

Lagrange Multipliers: 把 "有约束问题" 变成:

$$L(x, \lambda) \equiv f(x) + \lambda g(x)$$

Known as the Lagrangian function.

然后要求:

$$\nabla_x L(x, \lambda) = 0, \quad g(x) = 0$$

or just Lagrangian.

把带约束的优化问题, 重新表示成一个无约束的函数 (但引入了额外变量——乘子)

KKT 本质上是 Lagrange Multipliers 对不等式约束的扩展

问题: 为什么 Lagrange Multipliers 可以 "自动处理约束"?

切空间 tangent space
 对偶性 duality

梯度 = 法向量

The basic idea is to convert a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

Lagrangian 作为构造对偶性的核心工具.

(Duality)

Duality: 通过另一个视角 (对偶问题) 来估计原问题的最优值, 并找到这个估计的上限或下限.

Lagrangian 是工具, 用于构造对偶问题.

Second normal form: quadratic function

~~和~~ 和数据库的概念不是同一回事.

level set 定义: $L_c(f) = \{(x_1, \dots, x_n) \mid f(x_1, \dots, x_n) = c\}$

Differential topology/geometry

foundation: smooth manifolds

Morse lemma: 微积分 $\rightarrow$ 变分法 $\rightarrow$ 拓扑.

起点: 极值与二阶近似 = 为什么 critical point 附近函数 "像二次型"? Quadratic form

Quadratic approximations: The Taylor polynomial of order 2 generated by a twice-differentiable function $f(x)$ at $x=a$ is called the quadratic approximation of f at $x=a$.

linearization: Taylor polynomial of order 1.

题目: $f(x) = \ln(\cos x)$ $f(x) = e^{\sin x}$ $f(x) = 1/\sqrt{1-x^2}$ $f(x) = \cosh x$ $f(x) = \sin x$ $f(x) = \tan x$

= 二阶近似 本质上是用局部的二次函数 (quadratic form) 去替代复杂函数.

用途: 判断极值类型 (极小/极大/鞍点).

$$\left. \begin{aligned} f(x) &= \ln(\cos x) \\ f'(x) &= \frac{(\cos x)'}{\cos x} = \frac{-\sin x}{\cos x} = -\tan x & f'(0) &= 0 \\ f''(x) &= -(\sec^2 x) & f''(0) &= -1 \\ f(x) &\approx \frac{-1}{2}x^2 \end{aligned} \right\} = \text{二阶近似}$$

The careful analysis of non-degenerate (Hessian is non-degenerate) functions, so called Morse functions, is known as Morse Theory.

Morse Theory 最深的价值: 用局部二阶信息, 推断全局结构

(最神奇的地方, 只研究少数几个临界点, 就能知道整个空间长什么样).

Critical points: 临界点. 满足 $f'(x)=0$ 或 $\nabla f(x)=0$

题目: Does $f(x) = 2e^x/(1+x^2)$ have an absolute minimum value? An absolute maximum?

If so, find them or give reasons why they fail to exist. List all critical points of f .

高维法则求导

Quotient rule: $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$

$$f'(x) = \frac{2e^x(1+x^2) - 2e^x \cdot 2x}{(1+x^2)^2} = \frac{2e^x(x-1)^2}{(1+x^2)^2}$$

$$f'(x) = 0 \Rightarrow x = 1 \quad f'(x) \geq 0 \text{ 单调增}$$

$x=1$ 是 critical point. 但它并非

极值 (非极大非极小), 而是 flat stationary point.